

TYPES OF DATABASE:

1.3.1 Relational Databases (RDBMS)

Imagine your school keeps student information in **neat tables**, just like an Excel spreadsheet.

Roll No	Name	Class
1	Aisha	5A
2	Ravi	5B

Each table stores one type of information, and tables can be connected (related) using a common value, like Roll No.

For example:

- One table has **Students**
- Another table has **Marks**
- Roll No connects them together.

This is called a **Relational Database Management System (RDBMS)**.

Examples:

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server

Why use it?

- Data is organized and easy to find.
- Prevents duplicate data.
- Great for banks, schools, hospitals, and businesses.

Think of it as: A well-organized filing cabinet with labeled folders.

1.3.2 NoSQL Databases (MongoDB, Cassandra)

Sometimes data doesn't fit nicely into tables.

Imagine you have a toy box:

- One box has cars.
- Another has dolls.
- Another has LEGO.
- Each box can hold different things.

A **NoSQL database** stores data more flexibly than tables.

MongoDB

MongoDB stores information in **documents** (similar to JSON).

Example:

```
{
  "name": "Ravi",
  "age": 12,
  "hobbies": ["Cricket", "Drawing"]
}
```

Each person's information can be different.

Good for:

- Websites
- Mobile apps
- Social media

Cassandra

Cassandra is built for handling **huge amounts of data** across many computers.

Imagine a library with thousands of shelves in different cities. Even if one library closes, people can still borrow books from another.

Good for:

- Facebook-like apps
- Banking
- Online shopping
- Large companies

Why use NoSQL?

- Very fast.
- Can store different types of data.
- Can easily grow to handle millions of users.

Think of it as: A toy box where every compartment can store different kinds of toys.

1.3.3 Data Warehouses and Data Lakes

Both store **large amounts of data**, but they organize it differently.

Data Warehouse

Imagine a school library.

Every book:

- Has a proper label.
- Is placed on the correct shelf.
- Is easy to find.

A **Data Warehouse** stores **clean, organized data**.

Businesses use it to answer questions like:

- Which product sold the most?
- How much money did we earn this month?
- Which city has the most customers?

Characteristics

- Organized
- Structured
- Easy for reports and analysis

Think of it as: A neat library.

Data Lake

Now imagine a huge lake.

People throw in:

- Photos
- Videos
- Documents
- Audio files
- Sensor data
- Emails

Nothing is organized yet.

A **Data Lake** stores **all kinds of data**, whether organized or not.

Later, when someone needs the data, they can sort and analyze it.

Good for:

- Artificial Intelligence (AI)
- Big Data
- Machine Learning
- Research

Think of it as: A giant lake where everything is collected before sorting.

Quick Comparison

Feature	RDBMS	NoSQL	Data Warehouse	Data Lake
Data Format	Tables	Flexible (documents, key-value, etc.)	Organized tables	Any type of data
Structure	Fixed	Flexible	Structured	Structured + Unstructured
Best For	Banks, schools	Apps, social media	Business reports	AI, Big Data
Examples	MySQL, Oracle	MongoDB, Cassandra	Snowflake, Amazon Redshift	Hadoop, Amazon S3

Easy Way to Remember

- **RDBMS** = □ Excel sheets connected together.
- **MongoDB** = □ Flexible documents.
- **Cassandra** = □ Massive data spread across many computers.
- **Data Warehouse** = □ Organized library for reports.
- **Data Lake** = □ Big lake storing every type of data.